wave were to stretch something four meters long by, say, 10^{-20} meters, it would stretch something four kilometers long by a thousand times as much, 10^{-17} meters. So, the longer the span being monitored, the easier it is to detect a change in its length. To capitalize on this, the LIGO experimenters actually direct the laser beams to bounce back and forth between mirrors at opposite ends of each tube more than a hundred times on each run, increasing the roundtrip distance being monitored to about 800 kilometers per beam. With such clever tricks and engineering feats, LIGO should be able to detect any change in the tube lengths that exceeds a trillionth of the thickness of a human hair—a hundred millionth the size of an atom.

Oh, and there are actually two of these L-shaped devices. One is in Livingston, Louisiana, and the other is about 2,000 miles away in Hanford, Washington. If a gravity wave from some distant astrophysical hullabaloo rolls by earth, it should affect each detector identically, so any wave caught by one experiment had better also show up in the other. This is an important consistency check, since for all the precautions that have been taken to shield the detectors, the disturbances of everyday life (the rumble of a passing truck, the grinding of a chainsaw, the impact of a falling tree, and so on) could masquerade as gravitational waves. Requiring coincidence between distant detectors serves to rule out these false positives.

Researchers have also carefully calculated the gravitational wave frequencies—the number of peaks and troughs that should pass by their detector each second—that they expect to be produced by a range of astrophysical phenomena including supernova explosions, the rotational motion of nonspherical neutron stars, and collisions between black holes. Without this information the experimenters would be looking for a needle in a haystack; with it, they can focus the detectors on a sharply defined frequency band of physical interest. Curiously, the calculations reveal that some gravitational wave frequencies should be in the range of a few thousand cycles per second; if these were sound waves, they’d be right in the range of human audibility. Coalescing neutron stars would sound like a chirp with a rapidly rising pitch, while a pair of colliding black holes would mimic the trill of a sparrow that’s received a sharp blow to the chest. There’s a junglelike cacophony of gravitational waves